OBU Connection examples for 808 Protocol with Server

As soon as OBU power on, after booting it will try to connect the configured server, and register himself. During register OBU receives verification code from Server. OUB resend the verification code to server and server confirm the verification code. After verification, OBU responds to other commands. The connection sequence is as per below.

1. OBU send register request to Server.
2. Server Response to register request with verification code.
3. After receipt of server register response, OBU will send Authentication request to Server.
4. On reception of Authentication request from OBU, Server sends common response to the OBU.
5. After this authentication OBU will allow other data communication.

Below are the format and example for above process.

1. **Protocol format.**

Marker	Header	Message Body	Check Code	Marker
1 Byte Value 0x7E	12 Bytes	N Bytes	1 Byte	1 Byte Value 0x7E

Message ID	Message attributes	Device Phone No.	Message Sr. No.
2 Bytes (WORD)	2 Bytes (WORD)	6 Bytes. Number in BDC form	2 Bytes (WORD)

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		Sub contracting	Data encryption			Message body length									

Example of Header

Byte No.	Byte Value	Remarks
Byte 1 & 2 Message ID	0x0001	Hex value 0x0001 Message ID for OBU register request.
Byte 3 & 4 Message Attribute	0x002F	Data not encrypted. Message Body length is 0x2F (47 Bytes)
Byte 5 – 10 OBU Phone number	0x008291915608	6 Bytes BDC number. Phone number is 8291915608
Byte 11 & 12 Message Flow No.	0x0006	Message serial number. Starts from 00.

2. Register request from OBU to server

Message Format for register request from OBU to Server

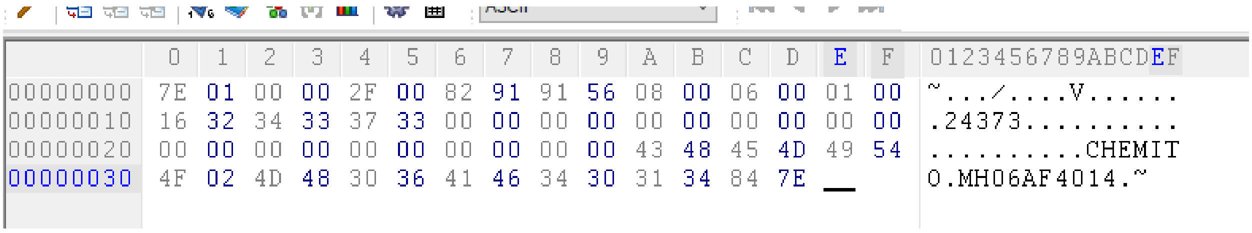
message definition.

Start Byte	Name	Data type	Description
0	Province ID	WORD	Marked the vehicle in which place(province), 0: reserved Server use the default ID.
2	City ID	WORD	Marked the vehicle in which City, 0: reserved, Server use the default ID.
4	Manufacturer ID	BYTE [5]	5 bytes manufacturer ID
9	Terminal ID	BYTE [20]	20 bytes this ID can be defined by manufacturer, if ID is less than 20 bytes ill the bit with 0x00.
29	CH OBU 7.0 ID	BYTE [7]	7 Bytes. It consists with capital letters and numbers; this ID can be defined by manufacturer.
36	Vehicle plate color	BYTE	Vehicle Plate Color, if no vehicle plate , the value is 0。
37	Vehicle flag	STRING	When vehicle plate color is 0, it means Vehicle VIN ; Or it means vehicle plate.

Example for Register request from OBU to Server

Byte No.	Byte Value	Remarks
Byte 0	0x7E	Marker
Byte 1 & 2	0x0100	Hex value 0x0100 Message ID for OBU register request.
Byte 3 & 4 Message Attribute	0x002F	Data not encrypted. Message Body length is 0x2F (47 Bytes)
Byte 5 – 10 OBU Phone number	0x008291915608	6 Bytes BDC number. Phone number is 8291915608
Byte 11 & 12 Message Flow No.	0x0006	Message serial number. Starts from 00.
Byte 13 & 14 Province ID	0x0001	Province ID set in OBU is 1
Byte 15 & 16 City ID	0x0016	City ID set in OBU is 16
Byte 17 – 21 Manufacturer ID	0x32, 0x34, 0x33, 0x37, 0x33	Manufacturer ID set in OBU ASCII value 24373
Byte 22 – 41 Terminal ID	0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00	Terminal ID. 20 bytes fixed. Presently not set
Byte 42 – 48 OBU ID (7 Bytes)	0x43, 0x48, 0x45, 0x4D, 0x49, 0x54, 0x4F	OBU ID set in OBU. ASCII value CHEMITO
Byte 49	0x02	Vehicle plate colour set in OBU.

Vehicle plate colour		0x00 = Null 0x01 = Blue 0x02 = Yellow
Byte 50 – 59 Vehicle flag	0x4D, 0x48, 0x30, 0x36, 0x41, 0x46, 0x34, 0x30, 0x31, 0x34	Vehicle flag n bytes. When Vehicle plate colour is 0 VIN will available. Otherwise Vehicle registration number will appears. ASCII text "MH06AF4014"
Byte 60	0x84	Check code. Check code calculated by 8 bit XOR all bytes from Byte 1 to Byte 59
Byte 61	0x7E	Marker



```
          0  1  2  3  4  5  6  7  8  9  A  B  C  D  E  F   0123456789ABCDEF
00000000  7E 01 00 00 2F 00 82 91 91 56 08 00 06 00 01 00  ~.../....V......
00000010  16 32 34 33 37 33 00 00 00 00 00 00 00 00 00 00  .24373..........
00000020  00 00 00 00 00 00 00 00 00 00 43 48 45 4D 49 54  ..........CHEMIT
00000030  4F 02 4D 48 30 36 41 46 34 30 31 34 84 7E        O.MH06AF4014.~
```

3. Server Response for Register request: After receiving the register request, server first checks Vehicle details and OBU/ MDVR in server data base and depend on data base server reply.

Start Byte	name	Data type	Description
0	Flow ID	WORD	Related MDVR register message flow ID
2	Result		0 : register success ; 1 : Vehicle Registered ; 2 : No vehicle data in data base 3 : MDVR registered ; 4 : No MDVR data in Database
3	Verification Code	STRING	This code only available after register successfully

Example for Register response from Server to OBU

Byte No.	Byte Value	Remarks
Byte 0	0x7E	Marker
Byte 1 & 2	0x8100	Hex value 0x8100 Message ID for OBU register request response
Byte 3 & 4 Message Attribute	0x001A	Data not encrypted. Message Body length is 0x1A (26 Bytes)
Byte 5 – 10 OBU Phone number	0x008291915608	6 Bytes BDC number. Phone number is 8291915608
Byte 11 & 12 Message Flow No.	0x0006	Message serial number. Starts from 00.
Byte 13 & 14 Response ID	0x0100	Response ID from Server to OBU. Send back Message ID received from OBU
Byte 15	0x00	Register status.

Result		0x00 = Register Success full 0x01 = Vehicle Register 0x02 = No vehicle data in server data base 0x03 = OBU/ MDVR Registered 0x04 = No OBU/ MDVR data in server data base
Byte 16 – 38 Verification code	0xD4, 0xC1, 0x4D, 0x48, 0x30, 0x36, 0x41, 0x46, 0x34, 0x30, 0x31, 0x34, 0x2D, 0x31, 0x32, 0x33, 0x34, 0x35, 0x36, 0x37, 0x38, 0x39, 0x30	Verification code generated by Server after confirming vehicle and OBU/ MDVR in data base. Verification code length is variable. Verification in example is "MH06AF4014-1234567890"
Byte 39 Check code	0x7C	Check code. Check code calculated by 8 bit XOR all bytes from Byte 1 to Byte 38
Byte 40	0x7E	Marker

7E8100001A008291915608000601000D4C14D48303641463430031342D31323334353637383930 7C7E

4. OBU request for Authentication to server: After receiving authentication (verification) code from server in register response, OBU request for authentication with Authentication (verification) code received from server.

Start Byte	name	Data type	Description
0	Verification code	STRING	After CH OBU 7.0 reconnected, then it will send the verification code

Example for sending verification code from OBU to server

Byte No.	Byte Value	Remarks
Byte 0	0x7E	Marker
Byte 1 & 2	0x0102	Hex value 0x0102 Message ID for OBU verification
Byte 3 & 4 Message Attribute	0x0017	Data not encrypted. Message Body length is 0x17 (23 Bytes)
Byte 5 – 10 OBU Phone number	0x008291915608	6 Bytes BDC number. Phone number in example is 8291915608
Byte 11 & 12 Message Flow No.	0x00F0	Message serial number. Starts from 00.
Byte 13 & 35 Response ID	0xD4, 0xC1, 0x4D, 0x48, 0x30, 0x36, 0x41, 0x46, 0x34, 0x30, 0x31, 0x34, 0x2D, 0x31, 0x32, 0x33, 0x34, 0x35,	Verification code received from Server. Verification in example is "MH06AF4014-1234567890"

	0x36, 0x37, 0x38, 0x39, 0x30	
Byte 36 Check code	0x04	Check code. Check code calculated by 8 bit XOR all bytes from Byte 1 to Byte 35
Byte 37	0x7E	Marker

```
          0  1  2  3  4  5  6  7  8  9  A  B  C  D  E  F   0123456789ABCDEF
00000000  7E 01 02 00 17 00 82 91 91 56 08 00 F0 D4 C1 4D  ~........V.....M
00000010  48 30 36 41 46 34 30 31 34 2D 31 32 33 34 35 36  H06AF4014-123456
00000020  37 38 39 30 04 7E                                7890.~
```

5. Server response for Authentication request to OBU: After receiving verification return request, server cross- check with the verification with generated verification code. Server response with common response if verification code matches.

Protocol Format for Server Common Response

Start Byte	Name	Type	Description
0	Marker	Byte	Packet start Marker
1	Message ID	Word	Message ID/ command for communication
3	Message Attribute	Word	Message attributes with message length
5	OBU Phone No.	6 Bytes	Phone number details register in OBU
11	Message Sr. No.	Word	Message serial number generated by OBU. Serial number starts from 00.
13	Answering Sr. No.	Word	Answering serial number generated by Server
15	Answering ID	Word	Answering ID, Message ID received from OBU has to revert here
17	Result/ Status	Byte	Confirmation result for OBU authentication. 0: Success/ Confirm 1: Failed 2: Error Message 3: Not available 4: Alarm
18	Check Code	Byte	Check code calculated by XOR of bytes 1 to 17
19	Marker	Byte	Packet End Marker

Protocol Example for server common response

Byte No.	Byte Value	Remarks

Byte 0	0x7E	Marker
Byte 1 & 2	0x8001	Hex value 0x8001 Message ID for Server common response
Byte 3 & 4 Message Attribute	0x0005	Data not encrypted. Message Body length is 0x05 (5 Bytes)
Byte 5 – 10 OBU Phone number	0x008291915608	6 Bytes BDC number. Phone number in example is 8291915608
Byte 11 & 12 Message Flow No.	0x0055	Message serial number. Starts from 00.
Byte 13 & 14 Response ID	0x0001	Answering Serial number from Server
Byte 15 & 16	0x0102	Message ID received from OBU in Authentication request.
Byte 17 Result	0x00	Verification result 0x00 = Successful/ Confirm 0x01 = Failed 0x02 = Error Message 0x03 = Not Available 0x04 = Alarm
Byte 18 Check code	0x0F	Check code. Check code calculated by 8 bit XOR all bytes from Byte 1 to Byte 17
Byte 19	0x7E	Marker

```
          0  1  2  3  4  5  6  7  8  9  A  B  C  D  E  F   0123456789ABCDEF
00000000  7E 80 01 00 05 00 82 91 91 56 08 00 55 00 01 01  ~.........V..U...
00000010  02 00 0F 7E                                      ...~
```

After OBU gets authenticated by Server, OBU starts sending location data and OBU heartbeat to Server at regular interval.

For location data and heartbeat server response is common server response.

Location data Message ID 0x0200

Message body format

Start Byte	Name	Type	Description
0	Marker	Byte	Packet start Marker
1	Message ID	Word	Message ID/ command for communication
3	Message Attribute	Word	Message attributes with message length
5	OBU Phone No.	6 Bytes	Phone number details register in OBU
11	Message Sr. No.	Word	Message serial number generated by OBU. Serial number starts from 00.

13	Alarm Sign	Double Word	The Alarm flag is defined in Table 24
17	Status	Double Word	The status bits are defined in Table 25
21	Latitude	Double Word	The Latitude value in degree, multiplied by 1000000.
25	Longitude	Double Word	The Longitude value in degree, multiplied by 1000000.
29	Elevation	Word	Altitude in meters
31	Speed	Word	1/10 Km/hour
33	Direction	Word	0-359, north is 0, clockwise
35	Time	BCD 6 Byte	YY-MM-DD-hh-mm-ss
40	Extension ID1	Byte	Value 1-255
41	Extension ID1 data Length	Byte	Extension information is provided in Table 27.
42	Extension ID1 data	N Bytes	Defined in Extension ID1 data length
42+N	Extension IDn	Byte	Value 1-255
42+N+1	Extension IDn data Length	Byte	Extension information is provided in Table 27.
42+N+2	Extension IDn data	M Bytes	Defined in Extension IDn data length
42+N+2+M	Check Code	Byte	Check code. Check code calculated by 8-bit XOR all bytes from Byte 1 to Byte 42+N+2+M
42+N+2+M+1	Marker	Byte	808 Protocol Marker

Table 24: Alarm Flag Definitions

Bit	Definition	Processing Instruction
0	Emergency alarm, triggered after touching the alarm switch	Cleared after receiving a response
1	Over Speed Alarm	The flag is maintained until the alarm condition is released.
2	Fatigue driving	The flag is maintained until the alarm condition is released. Applicable only if Fatigue system in installed.
3	Danger warning	Cleared after receiving a response.
4	GNSS module has failed	The flag is maintained until the alarm condition is released.
5	GNSS antenna is not connected or is cut	The flag is maintained until the alarm condition is released.
6	GNSS antenna is short circuit	The flag is maintained until the alarm condition is released.
7	OBU main power supply under voltage	The flag is maintained until the alarm condition is released.
8	OBU main power supply is powered off	The flag is maintained until the alarm condition is released.

9	BDC failure	The flag is maintained until the alarm condition is released.
10	TTS module failure	The flag is maintained until the alarm condition is released.
11	Camera failure	The flag is maintained until the alarm condition is released.
12	NA	NA
13	Speed warning	The flag is maintained until the alarm condition is released.
14	Fatigue driving warning	The flag is maintained until the alarm condition is released. Applicable only if fatigue system installed.
15	Reserved	
16	Reserved	
17	Reserved	
18	Accumulated driving timeout on the day	The flag is maintained until the alarm condition is released.
19	Over time parking	The flag is maintained until the alarm condition is released.
20	In and Out area	Cleared after receiving a response
21	In and out route	Cleared after receiving a response
22	Road section travel time is insufficient or too long	Cleared after receiving a response
23	Route deviation alarm	The flag is maintained until the alarm condition is released.
24	Vehicle VSS failure	The flag is maintained until the alarm condition is released.
25	Abnormal vehicle oil quality	The flag is maintained until the alarm condition is released.
26	The vehicle is stolen	The flag is maintained until the alarm condition is released.
27	Illegal ignition of the vehicle	Cleared after receiving response
28	Illegal displacement of the vehicle	Cleared after receiving response
29	Collision warning	The flag is maintained until the alarm condition is released.
30	Rollover warning	The flag is maintained until the alarm condition is released.
31	Illegal door opening alarm (When OBU does not set the area, it is not judge to open the door illegally)	Cleared after receiving response

Table 25 Status bit definition

Bit	Status
0	0: ACC off; 1: ACC on
1	0: Not positioned; 1: Positioned
2	0: North latitude; 1: South latitude
3	0: East longitude; 1: West longitude
4	0: Operational Status; 1: outage status
5	0: Latitude and longitude is not encrypted by the security plugin

	1: Latitude and longitude has been encrypted by the security plugin
6-7	Reserved
8-9	00: Empty Car; 01: Half loaded; 10: Reserved; 11: full loaded. Applicable if APC is installed
10	0: Vehicle oil path is normal; 1: Vehicle oil circuit is disconnected
11	0: Vehicle circuit is normal; 1: Vehicle circuit is abnormal
12	0: OBU door is unlocked; 1: OBU door is locked.
13	0: Door 1 off; 1: Door 1 on (Front Door)
14	0: Door 2 off; 1: Door 2 on (Middle Door)
15	0: Door 3 off; 1: Door 3 on (Back Door)
16	0: Door 4 off; 1: Door 4 on (Driver Door)
17	0: Door 5 off; 1: Door 5 on (Custom)
18	0: GPS satellite are not used for positioning 1: GPS satellite are used for positioning
19	NA
20	0: Positioning without GLONASS satellite; 1: Positioning with GLONASS satellite
21	0: Positioning without Galileo satellite; 1: Positioning with Galileo satellite
22-31	Reserved

Table: -26 Location Extension Item format

Field	Type of Data	Description and requirement
Extension ID	Byte	Value in the range 1-255
Extension ID data length	Byte	Extension ID data length
Extension ID data	N- Bytes	Extension ID data as per data length

Table: -27 Additional information definitions

Extension ID	Additional Information Length	Description and requirement
0x01	4 Byte	Mileage, 1/10 Km, corresponds to odometer
0x02	2 Byte	Oil quantity, WORD, 1/10L, corresponding to the fuel gauge
0x03	2 Byte	Speed acquired by the driving record function, WORD
0x04	2 Byte	Need to manually confirm the ID of the alarm event, WORD
0x05-0x10		Reserved
0x11	1 or 5 Byte	Additional information on the speeding alarm is shown in Table
0x12	6 Byte	Additional information on the entry/exit area/route alarm is
0x13	7 Byte	See section 30 for additional information on the travel time of
0x14-0x24		Reserved
0x25	4 Byte	Extended vehicle signal status bits, as defined in Table 31
0x2A	2 Byte	IO status bits, as defined in Table 32
0x2B	4 Byte	Analog, bit0-15, AD0; bit16-31, AD1.

0x30	1 Byte	BYTE, wireless communication network signal strength
0x31	1 Byte	BYTE, GNSS positioning satellite number
0xE0	Subsequent information length	Subsequent custom information length
0xE1-0xFF		Custom area

Table 28 over speed alarm additional information message body data format

Starting byte	Field	Data type	Description and requirements
0	Location type	BYTE	0: no specific location; 1: circular area; 2: rectangular area; 3: a polygonal area; 4: Road section
1	Area or segment ID	DWORD	If the location type is 0, there is no such field

Table 29 In and out area/route alarm additional information message body data format

Starting byte	Field	Data type	Description and requirements
0	Location type	BYTE	1: round area; 2: rectangular area; 3: a polygonal area; 4: route
1	Area or Line ID	DWORD	
5	Direction	BYTE	0: into; 1: Out

Table 30 Insufficient route travel time/excessive alarm additional message body data format

Starting byte	Field	Data type	Description and requirements
0	Segment ID	DWORD	
4	Road travel time	WORD	The unit is seconds (s)
6	Result	BYTE	0: insufficient; 1: too long

Table 31 Extended vehicle signal status bit

Bit	Definition
0	1: low beam signal
1	1: high beam signal
2	1: right turn signal
3	1: left turn signal
4	1: brake signal
5	1: reverse signal
6	1: fog light signal
7	1: position light
8	1: speaker signal

9	1: Air conditioning status
10	1: Neutral signal
11	1: retarder work
12	1: ABS work
13	1: heater work
14	1: clutch status
15-31	Reserved

Table 32: IO Status

Bit	Definition
0	1: deep sleep state
1	1: Sleep state
2-15	Reserved

Location Message Example

Byte Number	Byte Value	Description
0	0x7E	Packet start Marker
1	0x0200	Message ID
3	0x0044	Message attributes with message length
5	0x008291915608	Phone number details register in OBU
11	0x0178	Message serial number generated by OBU. Serial number starts from 00.
13	0x00000800	The Alarm flag is defined in Table 24
17	0x00040003	The status bits are defined in Table 25
21	0x0130C721	Latitude value in integer. Decimal value 19973921. After dividing it by 1000000. Final value is 19.973921
25	0x04662D68	Latitude value in integer. Decimal value 73805160. After dividing it by 1000000. Final value is 73.805160
29	0x0262	Altitude in meters 610 meters in example
31	0x0000	1/10 Km/hour Zero speed in example
33	0x00F7	0-359, north is 0, clockwise 247 degree in example
35	0x200414150132	14^{th} April 2020 and time 15:01:32 in example
40	0x01	Extension ID 0x01
41	0x04	Data length 4 byte
42	0x00000000	Presently Zero value in example
46	0x03	Extension ID 0x03
47	0x02	Data Length for Extension ID 3
48	0x0000	Presently Zero value in example
50	0x14	Extension ID 0x14
51	0x04	Data Length for Extension ID 0x14
52	0x00000001	Value 1 in example

56	0x15	Extension ID 0x15
57	0x04	Data Length for extension ID 0x15
58	0x00000003	Data value 3 in example
62	0x25	Extension ID 0x25
63	0x04	Data length for Extension ID 0x15
64	0x00000000	Data value zero in example
68	0x2B	Extension ID 0x2B
69	0x04	Data Length for Extension ID 0x2B
70	0x00000000	Data value zero in example
74	0x30	Extension ID 0x30
75	0x01	Data length for Extension ID 0x30
76	0x03	Data value in example Is 3
77	0x31	Extension ID 0x31
78	0x01	Data length for Extension ID 0x31
79	0x0C	Data value in example is 12 decimal
80	0x9A	Check code
81	0x7E	Marker

```
          0  1  2  3  4  5  6  7  8  9  A  B  C  D  E  F   0123456789ABCDEF
00000000  7E 02 00 00 44 00 82 91 91 56 08 01 78 00 00 08  ~...D....V..x...
00000010  00 00 04 00 03 01 30 C7 21 04 66 2D 68 02 62 00  .......0.!.f-h.b.
00000020  00 00 F7 20 04 14 15 01 32 01 04 00 00 00 00 03  ... .....2.......
00000030  02 00 00 14 04 00 00 00 01 15 04 00 00 00 03 25  ...............%
00000040  04 00 00 00 00 2B 04 00 00 00 00 30 01 03 31 01  .....+.....0..1.
00000050  0C 9A 7E                                         ..~
```

PIS status, Tamper alert and BDC alert upload

We adopt location information report message ID：0X0200 to upload Tamper alert and BDC alert and PIS status. This time the newly added protocol is additional information N, additional information ID: 0xEA, information length 2 bytes, 2 bytes additional content as below

Start Byte	Field	Data type	Description
00	Alert status	BYTE	bit0 is Tamper Alert; bit1 BDC Alert， 0：No
01	PIS status	BYTE	bit0-bit3 indicate in turn:PIS1->FDU,PIS2->RDU， PIS3->IDU， PIS4->SDU, 1: working status; 0: not working stat

The sample of uploading PIS status, Tamper alert and BDC alert as below

7E0200004901866498186400030000000100004000101C2A0E60658C7C00175000000002105261
51659010400000020403020000E4020320250400000000 2B04000000000300100310100170100180
400000000EA020000287E

808 Marker：7E
Message ID：0200
Message body attribute：0049
MDVR Mobile phone number（6Bytes）：018664981864
Message flow number, from 0 start（2Bytes）：0003
Message body content：

Alarm（4Bytes）：00000010
Status（4Bytes）：00040001
Latitude（4Bytes）： 01C2A0E6 29.532390 degree is multiplied by the 6th power of 10
Longitude（4Bytes）：0658C7C0 106.481600 degree is multiplied by the 6th power of 10
Elevation（2Bytes）：0175 373 Meters
Speed（2Bytes）：0000
Direction（2Bytes）：0000
Time（6Bytes）：20 05 20 20 06 58
Location extension information：
Extension information 1:
Extension information ID：01
The length of Extension information：04
The content of Extension information：00000204
Extension information 2:
Extension information ID：03
The length of Extension information：02
The content of Extension information：00000
Extension information 3:
Extension information ID：E4
The length of Extension information：02
The content of Extension information：0320
.
.
.
.
Extension information N:
Extension information ID：EA
The length of Extension information：02
The content of Extension information：0000
Check code：0x28
808 Marker：7E

Attendance Process:

A brief introduction to the attendance process: The OBU first attendance 0x0B05, upload driver information. After the Server receives 0x0B05, it searches the database according to the driver number. Compare, then the server first returns with Attendance Server response (Message ID 0x8B05), then send 0x8300.
The format of the Server receipt 0x8300: "(driver name), you sign in/ sign on the (Bus license plate) bus successfully.

For example: Naresh Panchal, you signed in the XX4014 bus successfully.

Driver Log in Protocol format

Start Byte	Name	Type	Description
0	Marker	Byte	Packet start Marker
1	Message ID	Word	Message ID/ command for communication

3	Message Attribute	Word	Message attributes with message length
5	OBU Phone No.	6 Bytes	Phone number details register in OBU
11	Message Sr. No.	Word	Message serial number generated by OBU. Serial number starts from 00.
13	Route Number	DWORD	Route number 4 Byte
17	Employee ID	STRING	Employee ID ASCII string. String termination by null byte. Maximum 256 Bytes
17+n	Time	6 Byte (BDC)	YY-MM-DD-hh-mm-ss. Attendance time
17+n+6	Attendance Type	Byte	See Table 31 attendance type
17+n+6+1	Attendance method	Byte	See Table 32 attendance method
17+n+6+1+1	Password	STRING	PASSWORD ASCII string. String termination by null byte. Maximum 256 Bytes
17+n+6+1+1+n	Check Code	Byte	Check code calculated by XOR of bytes 1 to 17+n+6+1+1+n
17+n+6+1+1+n+1	Marker	Byte	Packet End Marker

Table 31 Attendance Type

Name	Value
On Duty	0x01
Off Duty	0x02
Sign in	0x03
Sign Out	0x04
Check	0x05
Reserved	0x06 – 0x7F
Customized	0x80 – 0xFF

Table 32 Attendance Methods

Name	Value	Description
Employee Card Attendance	0x01	Employee swipe their attendance and take this value without distinguishing between employee categories
ID number attendance	0x02	Employee manually enter job number attendance, take this value when there is no need to distinguish employee categories
Driver attendance	0x03	Driver swap card attendance
Ticker attendance	0x04	Ticker swap card attendance
Reserved	0x05 – 0x7F	

Customized	0x80 – 0xFF	

Driver Login Protocol Example

Byte Number	Byte Value	Description
0	0x7E	Packet start Marker
1	0x0B05	Message ID for Attendance
3	0x0015	Message attributes with message length
5	0x008291915608	Phone number details register in OBU
11	0x0B9F	Message serial number generated by OBU. Serial number starts from 00.
13	0x0000002D	Route Number 4 Byte
17	0x65, 0x62, 0x30, 0x30, 0x30, 0x34, 0x66, 0x36, 0x00	Employee ID. In example ID eb0004f6
25	0x200415104613	BCD Date and time. 15th April 2020 and 10:46:13 time in example
31	0x04	Attendance type. In example Sign out.
32	0x01	Attendance mode. In example Sign out by Attendance card.
33	0x31,0x32,0x33,0x34,0x35,0x36,0x00	Password, In example 123456
40	0x5B	Check code
41	0x7E	Marker

7E0B050015008291915608 0B9F0000002D65623030303466360020041510461304013132333435360 05B7E

Server Response for Driver Attendance command format (Message ID 0x8B05)

Start Byte	Name	Type	Description
0	Marker	Byte	Packet start Marker
1	Message ID	Word	Message ID/ command for communication
3	Message Attribute	Word	Message attributes with message length
5	OBU Phone No.	6 Bytes	Phone number details register in OBU
11	Message Sr. No.	Word	Message serial number generated by OBU. Serial number starts from 00.
13	Attendance Response	BYTE	Byte details as per Table 60
14	Time	6 Byte (BCD)	Response date and time. Format YY-MM-DD-hh-mm-ss.
20	Attendance response string	STRING	Attendance response string, terminated by NULL byte
20 + n	Check Code	Byte	Check code calculated by XOR of bytes 1 to 20 + n
21 + n	Marker	Byte	Packet End Marker

Table 60: Attendance response

Byte Value	Description
0x00	Invalid Card
0x01	On Duty
0x02	Off Duty
0x03	Check In or Log in
0x04	Check Out or Log off
0x05	Routine Inspection
0x06 – 0xFF	Reserved

Server Response for Driver Attendance command (Message ID 0x8B05)

Byte No.	Byte Value	Remarks
Byte 0	0x7E	Marker
Byte 1 & 2	0x8B05	Hex value 0x8B05 Message ID for Server common response
Byte 3 & 4 Message Attribute	0x000E	Data not encrypted. Message Body length is 0x0E (14 Bytes)
Byte 5 – 10 OBU Phone number	0x008291915608	6 Bytes BDC number. Phone number in example is 8291915608
Byte 11 & 12 Message Flow No.	0x0000	Message serial number. Starts from 00.
Byte 13 Attendance Response	0x03	Attendance Response as per Table 60. Driver Sign in shown in example.
Byte 14 – 19 Attendance response Date & Time	0x20, 0x04, 0x20, 0x14, 0x10, 0x24	Driver Attendance response Date and Time
Byte 20 – 26 Response String	0x4E, 0x41, 0x52, 0x45, 0x53, 0x48, 0x00	Attendance response string, terminated by NULL byte. Driver Name in example "NARESH"
Byte 27 Check code	0x78	Check code. Check code calculated by 8 bit XOR all bytes from Byte 1 to Byte 26
Byte 28	0x7E	Marker

```
          0  1  2  3  4  5  6  7  8  9  A  B  C  D  E  F   0123456789ABCDEF
00000000  7E 8B 05 00 0E 00 82 91 91 56 08 00 00 03 20 04  ~........V.... .
00000010  20 14 10 24 4E 41 52 45 53 48 00 78 7E            ..$NARESH.x~_
```

After sending above response, Server sends text message to OBU. Text Message format for Attendance.

Text Message Format (0x8300)

Start Byte	Name	Type	Description
0	Marker	Byte	Packet start Marker
1	Response Message ID	Word	Message ID/ command for communication
3	Message Attribute	Word	Message attributes with message length
5	OBU Phone No.	6 Bytes	Phone number details register in OBU

11	Message Sr. No.	Word	Message serial number generated by OBU. Serial number starts from 00.
13	Attendance Marker	Byte	Attendance Marker as per Table 38 text information flag
14	Text Message	String	Attendance Text message. Terminated by NULL byte
14+n+1	Check Code	Byte	Check code calculated by XOR of bytes 1 to 14+n+1
14+n+1+1	Marker	Byte	Packet End Marker

Table 38 Text information Flag Meaning

BIT	SIGN
0	1: Emergency
1	Reserved
2	1: OBU Display
3	1: OBU TTS Broadcast
4	1: Advertising screen display
5	0: Centre Navigation information; 1: CAN fault code information
6-7	Reserved

Example for Attendance response from Server to OBU (Message ID 0x8300)

Byte No.	Byte Value	Remarks
Byte 0	0x7E	Marker
Byte 1 & 2	0x8300	Hex value 0x8300 Message ID for Server Attendance ID
Byte 3 & 4 Message Attribute	0x002A	Data not encrypted. Message Body length is 0x002A (42 Bytes)
Byte 5 – 10 OBU Phone number	0x008291915608	6 Bytes BDC number. Phone number in example is 8291915608
Byte 11 & 12 Message Flow No.	0x0B9F	Message serial number. Starts from 00.
Byte 13 Attendance Marker	0x0C	Attendance Marker. In example shows OBU Display and OBU TTS broadcast.
Byte 15 – 54	0x4E, 0x61, 0x72, 0x65, 0x73, 0x68, 0x20, 0x79, 0x6F, 0x75, 0x20, 0x73, 0x69, 0x67, 0x6E, 0x20, 0x69, 0x6E, 0x20, 0x6F, 0x6E, 0x20, 0x58, 0x58, 0x34, 0x30, 0x31, 0x34, 0x20, 0x73, 0x75, 0x63, 0x63, 0x65, 0x73, 0x73, 0x66, 0x75, 0x6C, 0x6C, 0x79	Attendance text Message. In example “Naresh you sign in on XX4014 successfully”

Byte 55 Check code	0xB0	Check code. Check code calculated by 8 bit XOR all bytes from Byte 1 to Byte 54
Byte 56	0x7E	Marker


```
          0  1  2  3  4  5  6  7  8  9  A  B  C  D  E  F   0123456789ABCDEF
00000000  7E 83 00 00 2A 00 82 91 91 56 08 0B 9F 0C 4E 61  ~...*....V....Na
00000010  72 65 73 68 20 79 6F 75 20 73 69 67 6E 20 69 6E  resh you sign in
00000020  20 6F 6E 20 58 58 34 30 31 34 20 73 75 63 63 65   on XX4014 succe
00000030  73 73 66 75 6C 6C 79 B0 7E                       ssfully.~
```


After driver attendance, driver request for Operation, OBU send operation request to Server.

Operation request format (Message ID 0x0B09)

Start Byte	Name	Type	Description
0	Marker	Byte	Packet start Marker
1	Response Message ID	Word	Message ID/ command for communication
3	Message Attribute	Word	Message attributes with message length
5	OBU Phone No.	6 Bytes	Phone number details register in OBU
11	Message Sr. No.	Word	Message serial number generated by OBU. Serial number starts from 00.
13	Route Number	DWORD	4 Bytes
17	Employee Number	String	Employee number terminated by NULL byte. Max length 256 Bytes
17 + m	Operation request code	BYTE	See Table 42, Operation request code
18+m	Time	BCD[6]	Operation request data and time
24+m	Check Code	Byte	Check code calculated by XOR of bytes 1 to 18+m
25+m	Marker	Byte	Packet End Marker

Table 42 Operation request code

Name	Value	Remarks
Scheduling Request	0x01	
Off-Duty request (switch Driver)	0x02	
Refuel Request	0x03	
Aerating Request	0x04	
Charging Request	0x05	
Exit Operation	0x06	
Start Manually	0x07	When system can’t start a function（Ex: No GPS signal），driver can send a request to management platform.

End Manually	0x08	When system can’t stop a function （Ex: GPS signal），driver can send a request to management platform
Charter Service Request	0x09	
Repair Request	0x0A	
Other Request	0x0B	
Start Operation	0x0C	
Request Intercom	0x0D	
Reserved	0x0E – 0x7F	
Customized	0x80 – 0xFF	

Operation request Example

Byte No.	Byte Value	Remarks
Byte 0	0x7E	Marker
Byte 1 & 2	0x0B09	Hex value 0x0B09 Message ID for Operation request
Byte 3 & 4 Message Attribute	0x0013	Data not encrypted. Message Body length is 0x0013 (19 Bytes)
Byte 5 – 10 OBU Phone number	0x008291915608	6 Bytes BDC number. Phone number in example is 8291915608
Byte 11 & 12 Message Flow No.	0x0001	Message serial number. Starts from 00.
Byte 13 – 16 Route Number	0x00000001	Route Number 1 shown in example
Byte 17 – 25 Employee ID/ Number	0x65, 0x62, 0x30, 0x30, 0x30, 0x34, 0x66, 0x36, 0x00	Employee ID. In example ID eb0004f6
Byte 26 Operation Request	0x01	Scheduling operation requested in example
Byte 27 – 32 Operation Request Date & Time	0x20, 0x04, 0x20, 0x10, 0x30, 0x00	Date 20th April 2020 and time 10:30:00 shown in example
Byte 33 Check code	0xBB	Check code. Check code calculated by 8 bit XOR all bytes from Byte 1 to Byte 32
Byte 34	0x7E	Marker

```
          0  1  2  3  4  5  6  7  8  9  A  B  C  D  E  F   0123456789ABCDEF
00000000  7E 0B 09 00 13 00 82 91 91 56 08 00 01 00 00 00   ~........V......
00000010  01 65 62 30 30 30 34 66 36 00 01 20 04 20 10 30   .eb0004f6.. . .0
00000020  BB 7E                                             .~_
```

Operation request Response Format from Server to OBU (Message ID 0x8B09)

Start Byte	Name	Type	Description
0	Marker	Byte	Packet start Marker
1	Response Message ID	Word	Message ID/ command for communication

3	Message Attribute	Word	Message attributes with message length
5	OBU Phone No.	6 Bytes	Phone number details register in OBU
11	Message Sr. No.	Word	Message serial number generated by OBU. Serial number starts from 00.
13	Code of Operation Request	WORD	Serial number of responded operation request
15	Result of Operation Request	BYTE	1: Agree; 0: Disagree
16	Respond Time	BCD[6]	Respond Date and Time
22	Departure Queue	BCD[6]	See Table 34
22+m	Additional Info	STRING	Additional Info string terminated by NULL Byte.
22+m+n	Check Code	Byte	Check code calculated by XOR of bytes 1 to 22+m+n
22+m+n+1	Marker	Byte	Packet End Marker

Table 34 Departure Queue

Start Byte	Name	Type	Description
0	Route Number	DWORD	Route Number
4	Road Sign	STRING	String Terminated by NULL string.
4+m	Planned Trip	STRING	A planned trip for the unique identification of the route schedule. Terminated by NULL byte
4+m+n	Bus Number	STRING	String terminated by NULL Byte
4+m+n+o	Business Type	BYTE	See Table 13
5+m+n+o	Schedule Type	BYTE	See Table 35
6+m+n+o	Driver Number	STRING	String terminated by NULL Byte. Max 50 Bytes
6+m+n+o+p	Driver Name	STRING	String terminated by NULL Byte.
6+m+n+o+p+q	Conductor 1 Number	STRING	String terminated by NULL Byte. Max 50 bytes
6+m+n+o+p+q+r	Conductor 2 Number	STRING	String terminated by NULL Byte. Max 50 bytes
6+m+n+o+p+q+r+s	Beginning time	BCD[6]	Beginning Date and Time
12+m+n+o+p+q+r+s	Ending time	BCD[6]	Ending date and time
18+m+n+o+p+q+r+s	Starting Station Number	DWORD	
22+m+n+o+p+q+r+s	Starting Station Name	STRING	String terminated by NULL Byte. Max 50 bytes.
22+m+n+o+p+q+r+s+t	Termination Station Number	DWORD	

26+m+n+o+p+q+r+s+t	Termination Station Name	STRING	String termination by NULL Byte. Max 50 Bytes

Table 13 Business Type

Name	Value
Upward Run	0x01
Downward Run	0x02
Circular Run	0x03
Stop at Origin station	0x04
Stop at Termination station	0x05
Reserved	0x06 – 0x1F
Customized	0x20 – 0x7F
Exit Station	0x80
Entry Station	0x81
Refuel	0x82
Inflate	0x83
Charging	0x84
Low level of Repair	0x85
High level of Repair	0x86
First level of Maintenance (low)	0x87
Second level of Maintenance	0x88
Third level of Maintenance (High)	0x89
No Passenger stay	0x8A
Stop at Station	0x8B
Cancel the Plan	0x8C
Reserved	0x8D – 0x9F
Customized	0xA0 – 0xFF

Table 35 Schedule Type

Name	Value
The whole road	0x01
Inter-regional	0x02
Stay at Station	0x03
Reserved	0x04 – 0x7F
Customized	0x80 – 0xFF

Operational Request response example

Byte No.	Byte Value	Remarks
Byte 0	0x7E	Marker
Byte 1 & 2	0x8B09	Hex value 0x8B09 Message ID for Operation request response
Byte 3 & 4 Message Attribute	0x007A	Data not encrypted. Message Body length is 0x007A (122 Bytes)
Byte 5 – 10 OBU Phone number	0x008291915608	6 Bytes BDC number. Phone number in example is 8291915608
Byte 11 & 12 Message Flow No.	0x0001	Message serial number. Starts from 00.

Byte 13 – 14 Serial Number	0x0001	Operation response Serial Number 1 shown in example
Byte 15 Result	0x01	Operation request response result. In example shown request Agree
Byte 16 Respond time	0x20, 0x04, 0x20, 0x10, 0x30, 0x00	Response Date and time. In example date 20^{th} April 2020 and time 10:30:00
Byte 22 Route Number	0x00, 0x00, 0x00, 0x64	Route number in integer. In example route number is 100.
Byte 26 Road sign	0x34, 0x35, 0x00	Road sign string. Terminated with NULL byte. 45 road sign is in example
Byte 29 Trip Number	0x36, 0x00	Trip Number string. Terminated by NULL Byte. In example trip number shown 6
Byte 31 Bus Number	0x58, 0x58, 0x34, 0x30, 0x31, 0x34, 0x00	Bus number string, terminated by NULL byte. In example Bus no. XX4014
Byte 38 Business Type	0x02	See Table 13. In example Downward run shown
Byte 39 Schedule Type	0x01	See Table 35. In example schedule shown for Complete route
Byte 40 Driver Number	0x00	Driver number string, terminated by NULL Byte. In example no driver number included
Byte 41 Driver Name	0x00	Driver name string, terminated by NULL byte. In example no driver name included
Byte 42 Conductor 1 Number	0x00	Conductor 1 number, terminated by NULL byte. In example no conductor number included
Byte 43 Conductor 1 Number	0x00	Conductor 2 number, terminated by NULL byte. In example no conductor number included
Byte 44 Start Time	0x20, 0x04, 0x20, 0x12, 0x00, 0x00	Start time and date. In example date 20^{th} April 2020 and Time 12:00:00
Byte 50 End Time	0x20, 0x04, 0x20, 0x13, 0x00, 0x00	End time and date. In example date 20^{th} April 2020 and Time 13:00:00
Byte 56 Starting Station No.	0x00, 0x00, 0x00, 0x01	Starting station number. In example starting station number is 1
Byte 60 Starting station Name	0x49, 0x2E, 0x53, 0x2E, 0x42, 0x2E, 0x54, 0x2E, 0x2D, 0x34, 0x33,0x00	Starting station string, terminated by NULL byte. In example starting station name is “I.S.B.T.-43”
Byte 72 Ending station No.	0x00, 0x00, 0x00, 0x12	Ending station number. In example ending station number is 18.
Byte 76 Ending station Name	0x49, 0x2E, 0x54, 0x2E, 0x20, 0x50, 0x61, 0x72, 0x6B, 0x00	Ending station name, terminated by NULL byte. In example station name is “I.T. Park”
Byte 86 Additional Contents	0x50, 0x6C, 0x65, 0x61, 0x73, 0x65, 0x20, 0x53, 0x74, 0x61, 0x72, 0x74, 0x20, 0x61, 0x74, 0x20, 0x31, 0x32, 0x20, 0x48, 0x6F, 0x75, 0x72, 0x73,	Additional content string, terminated by NULL byte. In example string is “Please Start at 12 Hours {"TS":3,"OTS":1,"OT":60}”

	0x20, 0x7B, 0x22, 0x54, 0x53, 0x22, 0x3A, 0x33, 0x2C, 0x22, 0x4F, 0x54, 0x53, 0x22, 0x3A, 0x31, 0x2C, 0x22, 0x4F, 0x54, 0x22, 0x3A, 0x36, 0x30, 0x7D, 0x00	
Byte 135 Check Code	0x--	Check code. Check code calculated by 8 bit XOR all bytes from Byte 1 to Byte 134
Byte 136	0x7E	Marker

OBU Request Driving plan to Server Example

Byte No.	Byte Value	Remarks
Byte 0	0x7E	Marker
Byte 1 & 2	0x0B07	Hex value 0x0B07 Message ID for Operation request response
Byte 3 & 4 Message Attribute	0x0004	Data not encrypted. Message Body length is 0x0004 (4 Bytes)
Byte 5 – 10 OBU Phone number	0x008291915608	6 Bytes BDC number. Phone number in example is 8291915608
Byte 11 & 12 Message Flow No.	0x0001	Message serial number. Starts from 00.
Byte 13 – 15 Operating Date	0x20, 0x04, 0x23	Operating date YY-MM-DD
Byte 16 Employee number	0x00	Employee Number, terminated by NULL byte. Max length 256 bytes
Byte 17 Check Code	0x--	Check code. Check code calculated by 8 bit XOR all bytes from Byte 1 to Byte 16
Byte 18	0x7E	Marker

Server send driving plan to OBU Example

Byte No.	Byte Value	Remarks
Byte 0	0x7E	Marker
Byte 1 & 2	0x8B07	Hex value 0x8B07 Message ID for Operation request response
Byte 3 & 4 Message Attribute	0x0090	Data not encrypted. Message Body length is 0x0090 (144 Bytes)
Byte 5 – 10 OBU Phone number	0x008291915608	6 Bytes BDC number. Phone number in example is 8291915608
Byte 11 & 12 Message Flow No.	0x0001	Message serial number. Starts from 00.
Byte 13 – 15 Operating date	0x20, 0x04, 0x23	Operating date YY-MM-DD
Byte 16 Running times	0x02	Bus running for 2 time
Byte 17 Road sign	0x31, 0x00	Road Sign String, terminated by NULL byte

Byte 19 Start Time	0x20, 0x04, 0x23, 0x10, 0x30, 0x00	Start time YY-MM-DD-hh-mm-ss
Byte 25 End Time	0x20, 0x04, 0x23, 0x11, 0x30, 0x00	End time YY-MM-DD-hh-mm-ss
Byte 31 Number of running plans	0x02	Bus running plan number
Byte 32 Route Number	0x00, 0x00, 0x00, 0x64	Route number in integer. In example route number is 100. **1st route in schedule**
Byte 36 Road sign	0x31, 0x00	Road sign string. Terminated with NULL byte. 1 road sign is in example
Byte 38 Trip Number	0x31, 0x00	Trip Number string. Terminated by NULL Byte. In example trip number shown 1
Byte 40 Bus Number	0x58, 0x58, 0x34, 0x30, 0x31, 0x34, 0x00	Bus number string, terminated by NULL byte. In example Bus no. XX4014
Byte 46 Business Type	0x02	See Table 13. In example Downward run shown
Byte 47 Schedule Type	0x01	See Table 35. In example schedule shown for Complete route
Byte 48 Driver Number	0x00	Driver number string, terminated by NULL Byte. In example no driver number included
Byte 49 Driver Name	0x00	Driver name string, terminated by NULL byte. In example no driver name included
Byte 50 Conductor 1 Number	0x00	Conductor 1 number, terminated by NULL byte. In example no conductor number included
Byte 51 Conductor 1 Number	0x00	Conductor 2 number, terminated by NULL byte. In example no conductor number included
Byte 52 Start Time	0x20, 0x04, 0x23, 0x12, 0x00, 0x00	Start time and date. In example date 23rd April 2020 and Time 12:00:00
Byte 58 End Time	0x20, 0x04, 0x23, 0x13, 0x00, 0x00	End time and date. In example date 23rd April 2020 and Time 13:00:00
Byte 64 Starting Station No.	0x00, 0x00, 0x00, 0x01	Starting station number. In example starting station number is 1
Byte 68 Starting station Name	0x49, 0x2E, 0x53, 0x2E, 0x42, 0x2E, 0x54, 0x2E, 0x2D, 0x34, 0x33,0x00	Starting station string, terminated by NULL byte. In example starting station name is "I.S.B.T.-43"
Byte 80 Ending station No.	0x00, 0x00, 0x00, 0x12	Ending station number. In example ending station number is 18.
Byte 84 Ending station Name	0x49, 0x2E, 0x54, 0x2E, 0x20, 0x50, 0x61, 0x72, 0x6B, 0x00	Ending station name, terminated by NULL byte. In example station name is "I.T. Park"
Byte 94 Route Number	0x00, 0x00, 0x00, 0x65	Route number in integer. In example route number is 101. **2nd Route in Schedule.**
Byte 98 Road sign	0x31, 0x00	Road sign string. Terminated with NULL byte. 1 road sign is in example
Byte 100 Trip Number	0x32, 0x00	Trip Number string. Terminated by NULL Byte. In example trip number shown 2

Byte 102 Bus Number	0x58, 0x58, 0x34, 0x30, 0x31, 0x34, 0x00	Bus number string, terminated by NULL byte. In example Bus no. XX4014
Byte 109 Business Type	0x02	See Table 13. In example Downward run shown
Byte 110 Schedule Type	0x01	See Table 35. In example schedule shown for Complete route
Byte 111 Driver Number	0x00	Driver number string, terminated by NULL Byte. In example no driver number included
Byte 112 Driver Name	0x00	Driver name string, terminated by NULL byte. In example no driver name included
Byte 113 Conductor 1 Number	0x00	Conductor 1 number, terminated by NULL byte. In example no conductor number included
Byte 114 Conductor 1 Number	0x00	Conductor 2 number, terminated by NULL byte. In example no conductor number included
Byte 115 Start Time	0x20, 0x04, 0x23, 0x13, 0x30, 0x00	Start time and date. In example date 23rd April 2020 and Time 13:30:00
Byte 121 End Time	0x20, 0x04, 0x23, 0x14, 0x30, 0x00	End time and date. In example date 23rd April 2020 and Time 14:30:00
Byte 127 Starting Station No.	0x00, 0x00, 0x00, 0x01	Starting station number. In example starting station number is 1
Byte 131 Starting station Name	0x49, 0x2E, 0x54, 0x2E, 0x20, 0x50, 0x61, 0x72, 0x6B, 0x00	Starting station string, terminated by NULL byte. In example starting station name is “I.T. Park”
Byte 141 Ending station No.	0x00, 0x00, 0x00, 0x12	Ending station number. In example ending station number is 18.
Byte 145 Ending station Name	0x49, 0x2E, 0x53, 0x2E, 0x42, 0x2E, 0x54, 0x2E, 0x2D, 0x34, 0x33,0x00	Ending station name, terminated by NULL byte. In example station name is “I.S.B.T.-43”
Byte 157 Additional Contents	0x00	Additional content string, terminated by NULL byte.
Byte 158 Check Code	0x--	Check code. Check code calculated by 8 bit XOR all bytes from Byte 1 to Byte 157
Byte 159	0x7E	Marker

CAN BUS data upload:
message ID 0x0705 is used to upload CAN BUS data.

Table 1: The data format of “CAN bus data upload”

Start Byte	field	Data type	Descriptions and requirements
0	The number of data	WORD	the number of CAN bus data included, which is more than 0

2	CAN bus data receiving time	BCD[5]	The receiving time of the first CAN bus data with format of hh-mm-ss-msms
8	CAN bus data item		the definition is given in Table 2

Table 2: The data format of CAN

Start Byte	field	Data type	Descriptions and requirements
0	The number of data	WORD	the number of CAN bus data included, which is more than 0
2	CAN bus data receiving time	BCD[5]	The receiving time of the first CAN bus data with format of hh-mm-ss-msms
8	CAN bus data item		the definition is given in Table 2

The example of CAN BUS data upload:

7E0705001F01866498186400DF000216284900000CF00400000102030405060718FEF100000102
03040506075F7E

Byte No.	Byte Value	Remarks
Byte 0	0x7E	Marker
Byte 1 & 2	0x0705	Hex value 0x0705 Message ID for can bus data.
Byte 3 & 4 Message Attribute	0x001F	Data not encrypted. Message Body length is 0x001F. It will vary accordingly can parameters count.
Byte 5 – 10 OBU Phone number	0x018664981864	6 Bytes BDC number. Phone number is 018664981864
Byte 11 & 12 Message Flow No.	0x0105	Message serial number. Starts from 00.
Byte 13	0x02	The number of CAN data item. Here in example is 02
Byte 14 - 18 City ID	0x1505450000	CAN data receiving time
Byte 19 – 22 CAN ID 1	0x0CF00400	CAN ID 1, its length is 4 byte
Byte 23 – 30 CAN ID 1 data	0x0001020304050607	CAN ID 1 data, its length is 8 byte
Byte 31 – 34 CAN ID 2	0x18FEF100	CAN ID 2
Byte 35 – 42 CAN ID 2 data	0x0001020304050607	CAN ID 2 data, its length is 8 byte
Byte 43	0xA6	Check code. Check code calculated by 8 bit XOR all bytes from Byte 1 to Byte 59
Byte 44	0x7E	Marker